Lesson plan · for the teacher

Nearest neighbours

Data science · Level 3 · Lesson 3

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Tell labels from numbers.
- Classify by nearest neighbours.
- Choose k sensibly.

Activity: The mystery fruit

1. Warm-up (5 min): أيّ مجموعة؟
2. Teach (8 min): Plot a few labelled dots, place a new one and circle its three nearest neighbours.
3. Measure (15 min): Groups weigh fruit, score colour and add their dots to the board.
4. Classify (7 min): Groups classify mystery fruits with k = 1 and k = 3 and compare.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- When did k change the answer?
- Why rescale grams?
- Who should make the final choice for a pupil?

Exit ticket

- Neighbours: blue, red, blue, green, blue. With k = 5, which label wins?

Try it in class tomorrow
Weigh some fruit, score their colour from 1 to 5 and plot them on the board. Bring a mystery fruit and let pupils find its three nearest neighbours.

Worksheet · for pupils

Nearest neighbours

Data science · Level 3 · Lesson 3

Name _____ Class _____ Date _____

1 **A model chooses a reading group for a new pupil. Does it predict a number or a label?**

◯ A number ◯ A label ◯ Neither

2 **Match each task to the kind of model.**

Predict a reading mark	• • • •	Regression
Choose a reading group	• • • •	Classification
Predict minutes of homework	• • • •	Regression

3 **Find the pupil closest to the new pupil (speed 5, accuracy 7) and print that pupil's group.**

```
pupils = [(2, 3, "red"), (6, 8, "blue"), (9, 2, "green"), (4, 5, "red")]
best = pupils[0]
print(best[2])
```

Write your program:

4 **Why is k = 1 risky?**

◯ One odd example can decide alone ◯ It is too slow ◯ It cannot give a label

5 **If one measure is in grams and the other is 1 to 5, the grams can decide almost everything.**

◯ True ◯ False

✎ **Where do you already sort pupils by comparing them with similar pupils you have taught? What could go wrong?**